

Sayf al Din Elkousy

Bellaire, TX | (713) 416-3107 | sayfaldin.elkousy@duke.edu | [LinkedIn](#) | [GitHub](#)

EDUCATION

Duke University

GPA: N/A

Double Major in Computer Science and Biology

Expected May 2029

- Relevant HS Coursework: AP CS Principles, Multivariable Calculus, AP Calc BC, AP Biology, AP Physics, AP Chemistry
- Society: Catalyst Tech Society, Duke Business Behind Health, Duke Investment Club (Trainee), Duke Arab Student Club

EXPERIENCE

Research Intern

January 2025 - July 2025

Fondren Orthopedic Research Institute

Houston, TX

- Co-authored a case study on calcific tendinopathy of the proximal hamstring in orthopedic patients, published in the National Institute of Health, by identifying statistical trends in the patient data and organizing the presentation of the study's findings.
- Demonstrated a 25% improvement in post-operative patient satisfaction among hip and knee replacement patients by analyzing a full year of single-hospital outcomes data on structured pre-operative patient education, and co-authored the resulting publication.
- Taught 60+ University of Houston medical students by designing a presentation on traumatic and degenerative bone lesions.
- Co-authored a 162-patient study in *Seminars in Arthroplasty: JSES* finding Traditional Medicare patients were 79% less likely to reach meaningful improvement after shoulder replacement, by gathering patient data from the hospital and co-writing the abstract.

Volunteer & Community Organizer

September 2023- August 2026

Fajr Scientific

Houston, TX

- Enabled 300+ surgeries across two medical missions to conflict zones by organizing patient data, surgical schedules, equipment.
- Helped raise \$2M for rebuilding infrastructure in war zones as a member of Fajr Scientific's fundraising committee for two major events, managing event logistics, preparing event venues, and supporting featured guest speakers before and during each event.
- Prepared presentation on ACL injury recovery delivered to 100+ physicians from Pakistan Orthopedic Association during medical mission to Pakistan, helping inform post-operative rehabilitation plans for 100+ patients who received surgery during the mission.
- Published a paper evaluating the utility of sports medicine within medical missions to underserved and conflict-affected regions.

Officer of the Youth Leaders Branch and Volunteer Branch

September 2024- August 2026

Egyptian American Society (EAS)

Houston, TX

- Served 1,000+ meals and 200+ care packages by coordinating five food distributions, including with Lunches of Love initiative, and leading 20-person volunteer teams; also organized and led Ramadan food drive that donated 200 meals to a local mosque.
- Planned, coordinated, volunteered at EAS Youth Group events for 75+ members, including senior potluck gathering and the EAS physicians network conference, helped distribute gifts to 30+ local refugee families in Houston during major religious holidays.

Assistant Coach

October 2023- August 2026

AMAANAH Refugee Organization

Houston, TX

- Provided free soccer instruction and equipment to 100+ students at low-income schools by organizing and co-leading a training initiative with AMAANAH soccer team, and organized and ran a free soccer clinic for 30+ children at the Refugee Day Festival.
- Helped the 40+ player AMAANAH refugee men's soccer team earn promotion to a higher league as an assistant coach by running practice sessions, and represented AMAANAH at Houston Dynamo MLS games as ball boy.

Starter & Captain

September 2021- June 2026

Bellaire High School Varsity Soccer / Rise Soccer Club ECNL/MLS NEXT

Houston, TX

- Ranked #19 in TX (Boys 2008) by PrepSoccer; 2023 MLS NEXT All-Star; 4x U.S. Youth National Team trial invitee
- Earned First Team All-District (2024–25, 2025–26) and All-Academic Team (2025–26); won 6A HISD District title (2025–26).

Founder & President

September 2024- June 2026

Sports Science Club

Houston, TX

- Founded a student sports science club and grew to 60 members by recruiting professional weightlifters and sports trainers as guest speakers, who delivered lectures and hands-on athletic training sessions to audiences of 40+ students at club meetings and events.
- Raised \$300+ for Challenged Athletes Foundation's adaptive athletics programs by organizing a pickleball tournament and socials.

PROJECTS

Particle Swarm Optimization in Quantum Engineering | *Python, NumPy*

QuERY Engineering Program (Quantum Engineering)

- Built a Particle Swarm Optimization simulation that models 100+ particles converging on a solution, demonstrating PSO as a heuristic, derivative-free approach to quantum engineering problems not suited to calculus-based optimization.
- Presented swarm convergence results to 50+ attendees at symposium of seven-week program mentored by Harvard/MIT students.

TECHNICAL SKILLS

Languages: Python, Java, HTML, CSS

Frameworks/Libraries: NumPy, Pandas, PyTorch

Developer Tools: Google Colab, Video Editing Software, Git, Claude Code